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During step descent, older adults exhibit decreased knee range of motion and increased vastus lateralis muscle activity

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ABSTRACT

Purpose: There is substantial evidence indicating an increase in the incidence of lower limb joint osteoarthritis with increasing age. One factor that may contribute to this is an age related reduction in the ability to attenuate the impulse generated during gait and step descent.
Scope: The aim of this study was to investigate the differences between older and younger adults, in the strategies used to attenuate the force generated between initial contact (IC) and maximum vertical ground reaction force (vGRF) (impulse phase) when descending a step.
Methods: Ten older participants aged 60–80 years (mean 65.3, SD ±5) and 10 younger participants aged 20–30 years (mean 22.8, SD ±2.5) took part in the study. Vertical ground reaction force (vGRF) and the time taken to reach maximum vGRF were measured. The maximum joint range of motion of the hip and knee during step descent was measured between IC and the maximum vGRF. Electromyography (EMG) was measured from four lower limb muscles and normalised to task for each individual.
Results: There was a statistically significant difference between older and younger adults in the amount of knee flexion and vastus lateralis EMG activity in the leading leg during the impulse phase of step descent. Older adults had less knee flexion ($F(1,18) = 5.48$; $p = 0.031$) and more vastus lateralis EMG activity ($F(1,18) = 5.21$; $p = 0.035$) during step descent than younger adults.
Conclusion: Older and younger adults used different strategies during step descent. Older adults demonstrated two strategies that have the potential to increase the impulse of a step.
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1. Introduction

Muscle weakness is a well-recognised secondary impairment in people with osteoarthritis [1–3] but recent research has also implicated it as a predisposing factor in the pathogenesis of osteoarthritis, particularly in weight bearing joints [4–6]. The reduction in muscle strength that often occurs with increasing age, may affect the ability to attenuate impulse during daily activities [7] and increase the risk of joint damage from the cumulative effect of reduced shock absorption [8,9] in the stepping leg during step descent. One of the main strategies to control the impulse and act as a shock absorber [10,11] is muscle activity which is temporally matched to the task [12]. Gluteus medius and quadriceps femoris control hip adduction and knee flexion, respectively, and both have a role in reducing impulse during stepping in healthy adults [10,13].

In previous studies investigating the kinematic requirements of step descent, the emphasis has been on maximum range of knee flexion for the whole task. This usually occurs in the support leg, as it lowers the stepping leg and is when most muscle activation is required to control the movement [14]. The range of knee flexion used in the impulse phase of step descent is not reported. Lark et al. [15] compared maximum passive knee range of motion, with the range used by older adults whilst performing a step descent. Protopapadaki et al. [16] compared maximum knee range of motion required for ascent and descent in healthy adults. Riener et al. [17] and Livingston et al. [18] investigated the effect of different step heights on maximum range of knee motion in healthy young adults. None of these studies focused on knee flexion during the impulse phase.

The aim of our study was to ascertain the differences between older and younger age groups in joint range of motion, muscle activity and vertical ground reaction force (vGRF). The hip and knee were selected, as hip adduction and knee flexion have previously been identified as important in attenuation of impulse. Muscles investigated were gluteus medius, vastus lateralis, gastrocnemius medialis and tibialis anterior.

2. Methods

2.1. Participants

Ten older participants (60–80 years) and 10 younger participants (20–30 years) were recruited via posters placed on university notice boards and word of mouth.

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Fig. 1. Experimental area with participant descending the step on to the force plate.

Five males and five females from the older age group and two males and eight females from the younger age group volunteered to participate in the study. Inclusion criteria were the ability to walk independently with no walking aid or assistance and without pain; no diagnosed arthritis, cardiovascular system disease or neurological injury and the ability repeatedly to descend a step safely. The university ethics committee approved the study and all participants gave written informed consent prior to commencement.

2.2. Procedure

Age, height and weight of all participants were recorded. Throughout testing, participants were barefoot and wore shorts and a singlet top to allow reflective markers and EMG electrodes to be attached to the skin.

The experimental area consisted of a 5 m walkway with a 20 cm step placed at the beginning of the walkway, 5 cm behind a force plate (Fig. 1). Participants stood on the step and were instructed to step down with either their right or left foot then continue walking to the end of the walkway. After each trial, the participant returned to the step for the next trial. Each participant undertook 40 trials with blocked randomisation used to determine the order of right or left step descent, resulting in 20 right and 20 left step descents. Participants were offered the opportunity to rest after 10 trials and could chose to terminate testing at any point (Table 1).

2.3. Kinematic data

A nine-camera, three-dimensional motion analysis system with Qualisys® software was used to collect kinematic data (Qualisys motion capture systems, Qualisys Medical AB, Sweden). The cameras were calibrated prior to each session and positioned around the experimental area, to record three-dimensional movement of the lower limbs.

Retroreflective spherical markers (19 mm diameter) were attached to the participant's skin using double-sided tape, on bony landmarks on the pelvis and both lower limbs as follows; both acromion processes, bilateral ASIS, mid PSIS, T8, offset bilateral shin and thigh, bilateral medial and lateral knee joint line and ankle malleoli, bilateral mid-point between 2nd and 3rd metatarsals and bilateral calcaneus. Kinematic data were collected at a sampling rate of 240 Hz.

Prior to the calculation of joint kinematics, marker trajectories were smoothed using a second order Butterworth low pass filter, with a cut off frequency of 12 Hz, to eliminate the noise from movement artefacts. The data from an initial static trial were used to construct a 3D rigid link segment model, using the Visual 3D® software (C-Motion Inc., USA). The static model was applied to the movement files; hip adduction and knee flexion were calculated from IC to maximum vGRF.

Table 1
Participant characteristics.

	Older group, mean (SD)	Younger group, mean (SD)
Age (years)	65.3 (SD 5.0)	22.8 (SD 2.5)
Weight (kg)	67.6 (SD 9.37)	66.4 (SD 11.88)
Height (m)	1.67 (SD 0.06)	1.72 (SD 0.09)

2.4. Kinetic data

GRF data were recorded using an AMTI force plate (Advanced Mechanical Technology Inc. USA; Model OR6-7-2000). Prior to each session, the force plate was zeroed. During the study participants stepped on to the force plate from the step and GRF data was collected at a rate of 1200 Hz via a computer data acquisition system. Kinetic data were smoothed using a Butterworth low pass filter with a 70 Hz cut off frequency, to eliminate high frequency noise from the GRF data. Maximum vGRF between IC and prior to "push off" was calculated within the Visual 3D® software programme (Fig. 2).

2.5. Electromyography

An eight-channel analogue multiplex system (AMT-8 Bortec Biomedical Ltd., Calgary, AB, Canada) with bipolar, silver/silver chloride surface EMG electrodes (Norotrode 20) was used to collect EMG signals.

Skin preparation included dry shaving the skin directly over the muscle belly on each leg. The skin was lightly abraded using skin preparation liquid and cleaned with an alcohol wipe. Electrodes were placed over selected muscles of both legs, according to recommendations by Hermans et al. [19] and a ground electrode was placed on the left fibula head. Each EMG lead had a pre-amplifier taped close to the electrode to reduce the incidence of movement artefacts. Impedance was measured and accepted at a level below 5 kΩ. The electrode leads were attached to a central collection port, on a belt around the subjects' waist. This was attached to the analog multiplex system via a long, lightweight lead to enable unimpeded movement. The EMG signal was collected using a bandpass filter of 10–1000 Hz with a preamplifier to enhance signal collection. For each participant, maximum EMG amplitude was calculated from the smoothed rectified signal from all step descents for each leg and each muscle during the entire stance phase of the step descent from IC to toe off. The integral of the maximum signal found over this period was normalised to the period from initial contact to maximum vGRF and accepted as the maximum EMG amplitude for the task. The muscle activity from IC to maximum vGRF for each of the muscles was expressed as a percentage of the *maximum for task* [20].

2.6. Digital camera

To verify the collected motion data and allow off-line visual checking a digital video camera (Panasonic, USA) collected images in real time and was synchronised with the 9 camera system. Recordings were made of all the steps performed by all the participants.

2.7. Statistical analysis

Kinematic and kinetic data were initially checked and found to be normally distributed. An independent samples *t*-test was undertaken to compare the means of weight and height between the age groups, no significant difference between the age groups was found. The mean for each dependent variable was calculated for all participants, the group mean for the two age groups was calculated as a mean of the means. A general linear model repeated measures ANOVA design was used to test for differences within the age groups, i.e. comparing right and left legs and between the age groups for the dependent variables. An *F ratio* for each dependent variable was generated. An alpha level of $p \leq 0.05$ was set for all statistical analysis. The data was analysed using SPSS v16.0 for Windows statistical software package (SPSS, Chicago, USA).

3. Results

All participants completed 40 trials of the step down task. There were no withdrawals from the trials.

To explore within participant variability within trials, the coefficient of variation (COV) was calculated for each of the dependent variables.

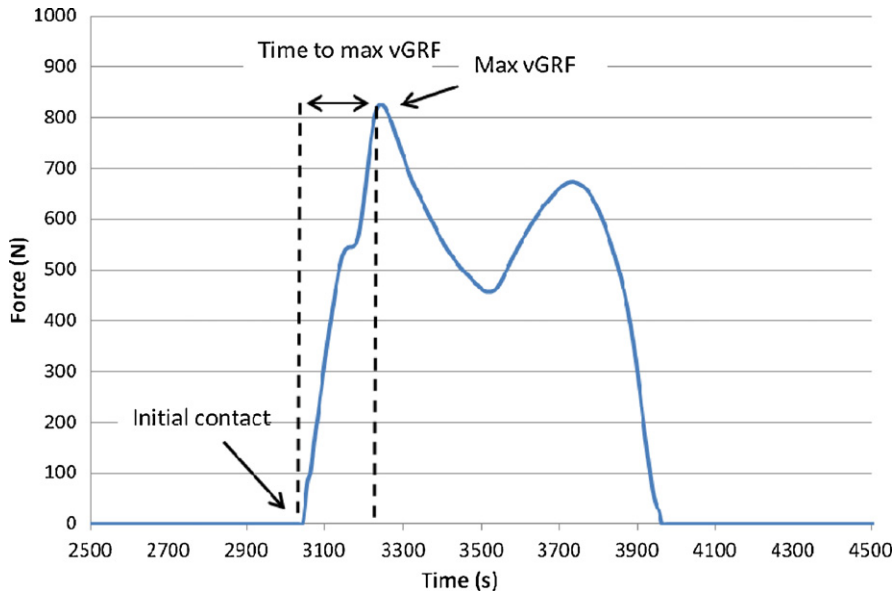

Fig. 2. Vertical GRF trace from force plate.

For muscle activity, the COV varied between 11% and 26% across participants which was similar in magnitude to COV of hip adduction and knee flexion (16–25% and 8–23%, respectively). Ground reaction forces showed the least variability within trials, 4–9% and 8–17% for maximum vGRF and time to maximum vGRF, respectively. Variability was not systematically associated with age. Higher variability was found where measurements were of a smaller magnitude (Table 2).

There were no differences found in the within group analysis, comparing right and left legs (except for gastrocnemius activity, which is considered below). The results for the between group analysis are reported below.

3.1. Hip adduction

The older participants had a mean maximum hip adduction angle of the left leg of 3.15° (SD ±2.75) and the right leg of 2.26° (SD ±2.58). The younger participants had a mean maximum hip adduction angle of the left leg of 4.75 (SD ±3.13) and the right leg of 5.36° (SD ±2.05). The amount of hip adduction was not statistically significantly different between younger and older participants ($F(1,18) = 2.74$; $p = 0.116$).

3.2. Knee flexion

The older participants had a mean maximum knee flexion angle of the left leg of 17.3° (SD ±7.8) and of the right leg 18.2° (SD ±6.1). The younger participants had a mean maximum knee flexion angle of the left leg of 25.5° (SD ±13.5) and of the right leg 27° (SD ±11.6). The difference between the two age groups was statistically significant with older participants using less knee flexion during the step descent task ($F(1,18) = 5.48$; $p = 0.031$).

3.3. Vertical ground reaction force

Data were normalised, according to participant's body weight (vGRF/BW).

The mean maximum vGRF/BW for older participants was, for the left leg 1.67 (SD ±0.3) and for the right leg 1.6 (SD ±0.22). For younger participants it was, for the left leg 1.58 (SD ±0.39) and for the right leg 1.58 (SD ±0.27). No statistically significant difference was found between younger and older participants for maximum vGRF/BW generated by the step descent task ($F(1,18) = 0.343$; $p = 0.565$).

Table 2
Ensemble mean and standard deviation values of dependent variables.

	Older				Younger			
	Left		Right		Left		Right	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Knee angle-sagittal plane (°)	17.30	7.82	18.23	6.11	25.55	13.5	27.01	11.59
Hip angle-frontal plane (°)	3.15	2.75	2.26	2.58	4.75	3.13	5.36	2.65
Max vGRF (N)	1124.66	357.72	1089.85	283.30	1021.20	234.23	1029.01	215.62
Max vGRF/BW	1.67	0.3	1.6	0.22	1.58	0.39	1.58	0.27
IC to max vGRF (s)	0.13	0.03	0.14	0.04	0.14	0.03	0.13	0.03
Gluteus medius[a]	45.3	9.9	40.9	16.7	46.6	7.7	45.4	8.6
Vastus lateralis[a]	55.1	8.9	50.8	10.3	46.7	8.6	44.3	6.2
Gastrocnemius[a]	43.49	7.8	29.67	19.9	38.1	7.6	31.2	12
Tibialis anterior[a]	35.7	15.4	27	11.4	36.5	10.7	26.8	12.8

IC, initial contact; vGRF, vertical ground reaction force; BW, body weight (N).
[a] All muscle activity is from IC to maximum vertical GRF (percentage of maximum for task).

The mean time from IC to maximum vGRF, for older adults for the left leg was 0.127 s (SD ±0.03) and the right leg was 0.138 s (SD ±0.04). For younger adults for the left leg it was 0.142 s (SD ±0.026) and the right leg 0.134 s (SD ±0.036). The lowest scores, corresponding to the fastest loading rates were found in the older group, but no significant difference was found between older and younger participants for the time to reach maximum vGRF ($F(1,18) = 0.268$; $p = 0.611$).

3.4. Gluteus medius

The older participants used a mean of 45.3% (SD ±9.9) of the maximum for task on the left leg and 40.9% (SD ±10.7) on the right. The younger participants used a mean of 46.6% (SD ±7.7) of the maximum for task on the left leg and 45.4% (SD ±8.6) on the right. There was no significant difference between older and younger participants in the amount of gluteus medius activity ($F(1,18) = 0.755$; $p = 0.396$).

3.5. Vastus lateralis

The older participants used a mean of 55.1% (SD ±8.9) of the maximum for task on the left leg and 50.8% (SD ±10.3) on the right. Younger participants used 46.7% (SD ±8.6) of the maximum for task on the left leg and 44.3% (SD ±6.2) on the right. There was a significant difference between the groups ($F = 5.21$; $p = 0.035$), with older participants using significantly more muscle activity than the younger participants (Fig. 3a and b).

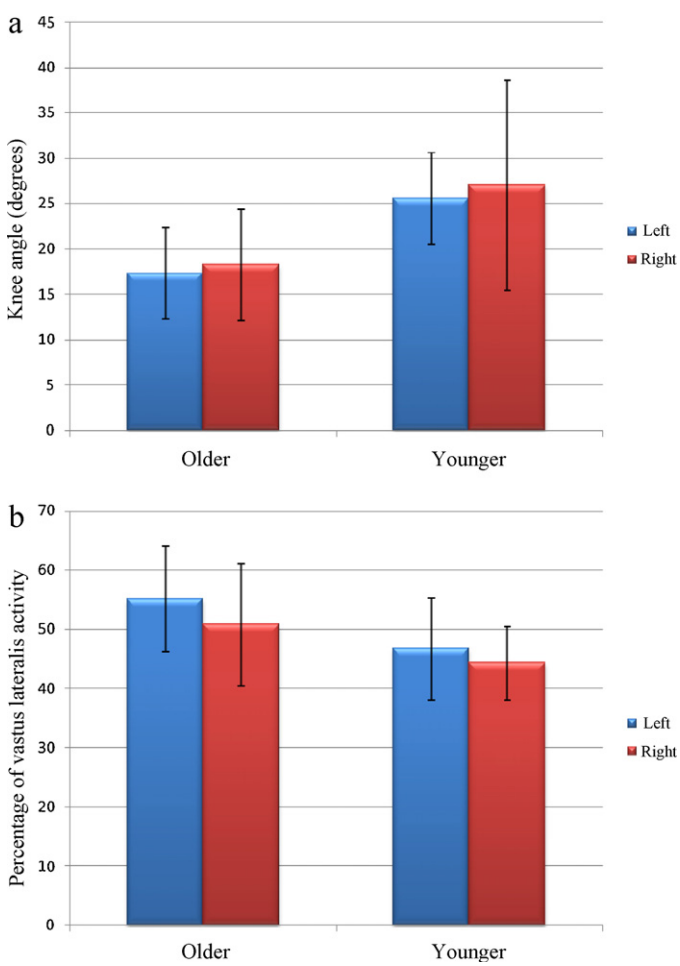

Fig. 3. (a) Mean maximum knee flexion angles between initial contact and maximum vertical GRF. (b) Vastus lateralis activity from initial contact to maximum vertical GFR.

3.6. Gastrocnemius

A main effect of leg (right compared with left) was found for gastrocnemius activity, so independent *t*-tests were used to analyse right and left legs separately. No significant difference was found between older and younger participants for either leg. Left gastrocnemius activity, $t = -1.73$, $p = 0.1$; right gastrocnemius activity, $t = 0.22$, $p = 0.8$.

3.7. Tibialis anterior

The older participants used a mean of 35.7% (SD ±15.4) of the maximum for task on the left and 27% (SD ±11.4) on the right. Younger participants used a mean of 30.5% (SD ±10.7) of the maximum for task on the left and 26.8% (SD ±12.8) on the right. There was no significant difference between older and younger participants in the amount of tibialis anterior muscle activity ($F(1,18) = 0.232$; $p = 0.636$).

4. Discussion

The results of this study show a statistically significant difference between older and younger participants in the range of motion and muscle activity at the knee joint during the impulse phase of the stepping leg during stair descent.

Range of motion measurement between IC and maximum vGRF showed a difference in knee flexion in the stepping leg between older and younger adults indicating the relative stiffening of the older adult's knee joint. The finding of a significant difference in activation of vastus lateralis between older and younger participants was congruent with previous research into stair descent and helps explain the finding of a reduced range of knee flexion. Other research has investigated timing of onset of quadriceps femoris prior to initial contact [21], and pre-activation of vastus lateralis and co-activation of biceps femoris during the gait cycle [11]. Both these studies found that older participants pre-activate and co-activate muscles in the stepping leg significantly more than younger participants. They suggested this may be due to age related changes in muscles [22] making them less able to contract as quickly when required. The only other study that sought to quantify the activity of vastus lateralis after IC was by Reeves et al. [23]. These authors showed no difference in the pattern of normalised EMG recordings from vastus lateralis between older and younger adults. However recent research suggests that due to reduced muscle strength, older participants work at a higher relative capacity than younger [14]. There has been recent research into the theory that muscle weakness and dysfunction, directly attributable to age related sarcopenia, predisposes older adults to cumulative joint damage [8]. There has been very little investigation into muscle activity during impulse attenuation and conflicting evidence about the magnitude and rate of loading of vGRF during step descent. Our study reports the relative magnitude of vastus lateralis activity in the early impulse-attenuating phase of step descent.

No significant difference was found in vGRF or time taken to reach maximum vGRF between older and younger participants, in our study. This is a very different finding to the study by Robbins et al. [8] who noticed large differences between older younger participants in vGRF during a step descent. In our study a functional task was undertaken, whereas Robbins et al. instructed participants to step down and remain balanced on one leg for a few seconds, this act of deceleration, halting forward momentum, may have contributed to the increase in vGRF seen in older participants. Our results were somewhat surprising, as our study did identify a decrease in knee flexion angle and an increase in vastus lateralis activity at IC in older adults. It may be that while both the

reduction in knee flexion and the increase in muscle activity in older adults are statistically significant, they were not sufficient to increase vGRF significantly. A limitation of this study is that we did not investigate kinematics at the ankle; this may have revealed changes that could have accounted for the vGRF findings. Despite not finding an increase in vGRF, the findings of our study show a decrease in knee flexion and an increase in vastus lateralis activity in the stepping leg, which will stiffen the leg and has the potential to reduce the ability to attenuate impulse. An older or less physically active population may have yielded different results. Future studies would benefit from subdividing the older age category [24], for example 60–69, 70–79 and differentiating between active and sedentary older adults. The physiological changes that occur in muscle with aging and inactivity [25–27] make it possible that older, more sedentary adults are less able to attenuate impulse than fit adults in their 60s.

Previous research supports the theory that the finding of decreased knee flexion and increased vastus lateralis activity are in response to reduced stability [28] and a need to co-contract lower limb muscles prior to impulse [29]. In that case, is this a strategy that can be returned closer to the normal strategy observed in younger adults with targeted intervention to improve stability? Exercise interventions for older adults are often focused on increasing maximum strength but there is some evidence that task specific training with an emphasis on timing and accurate scaling of sub maximal force, can lead to significant improvements in returning muscle activity to patterns normally found in younger adults [30]. In support of this suggestion Hsu et al. [11] recommend that a lower limb strengthening programme for older adults should include focus on the timing of muscle contraction. Exercises could be focused on improving the speed and accuracy of muscle activity during the impulse phase of step descent and gait. This may result in increased confidence, to utilise more range of motion and reduce the potential for damage to lower limb joints.

In conclusion, this investigation into the differences in strategy used to descend from a step, has revealed both decreased joint range of motion and increased muscle activity in the knees of older versus younger participants. Although this study did not find a resultant increase in vGRF, both strategies have the potential to increase impulsive loading because they "stiffen" the knee and reduce the ability to absorb the impulse of landing. Investigation including older or less physically active adults may clarify if this is a major contributor to the progressive joint deterioration seen in older adults.

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Conflict of interest statement

No conflict of interest exists in regards to this study.

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